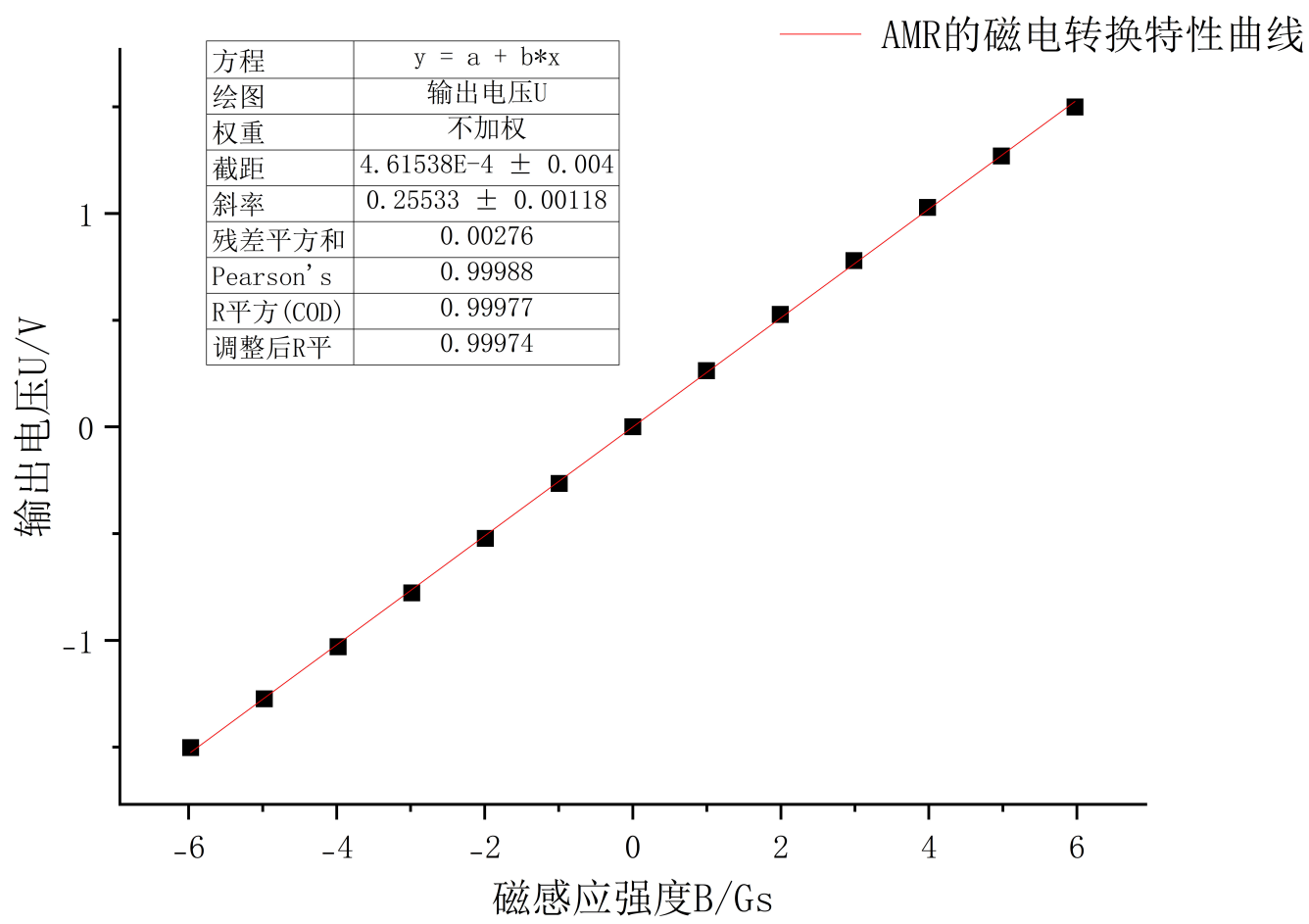
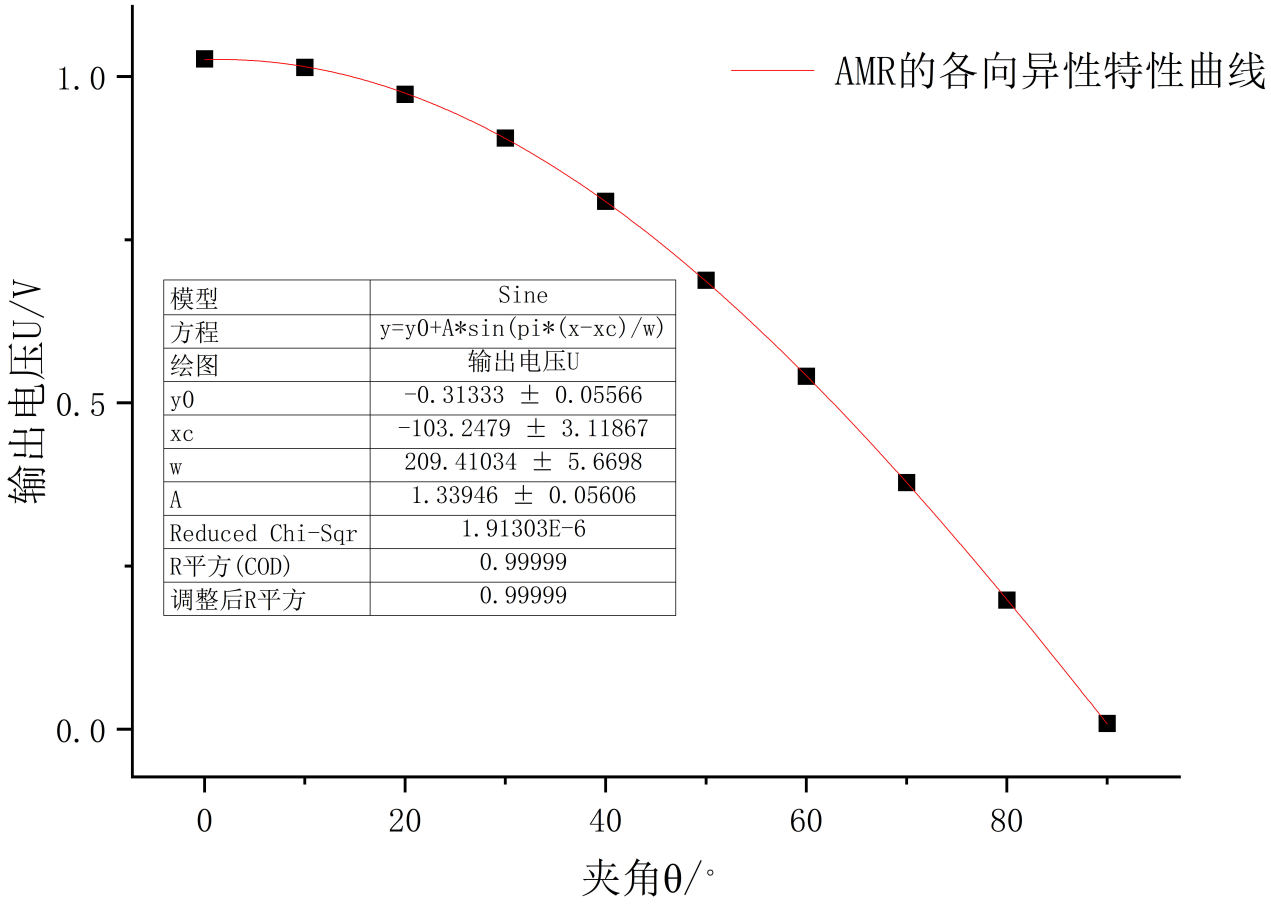
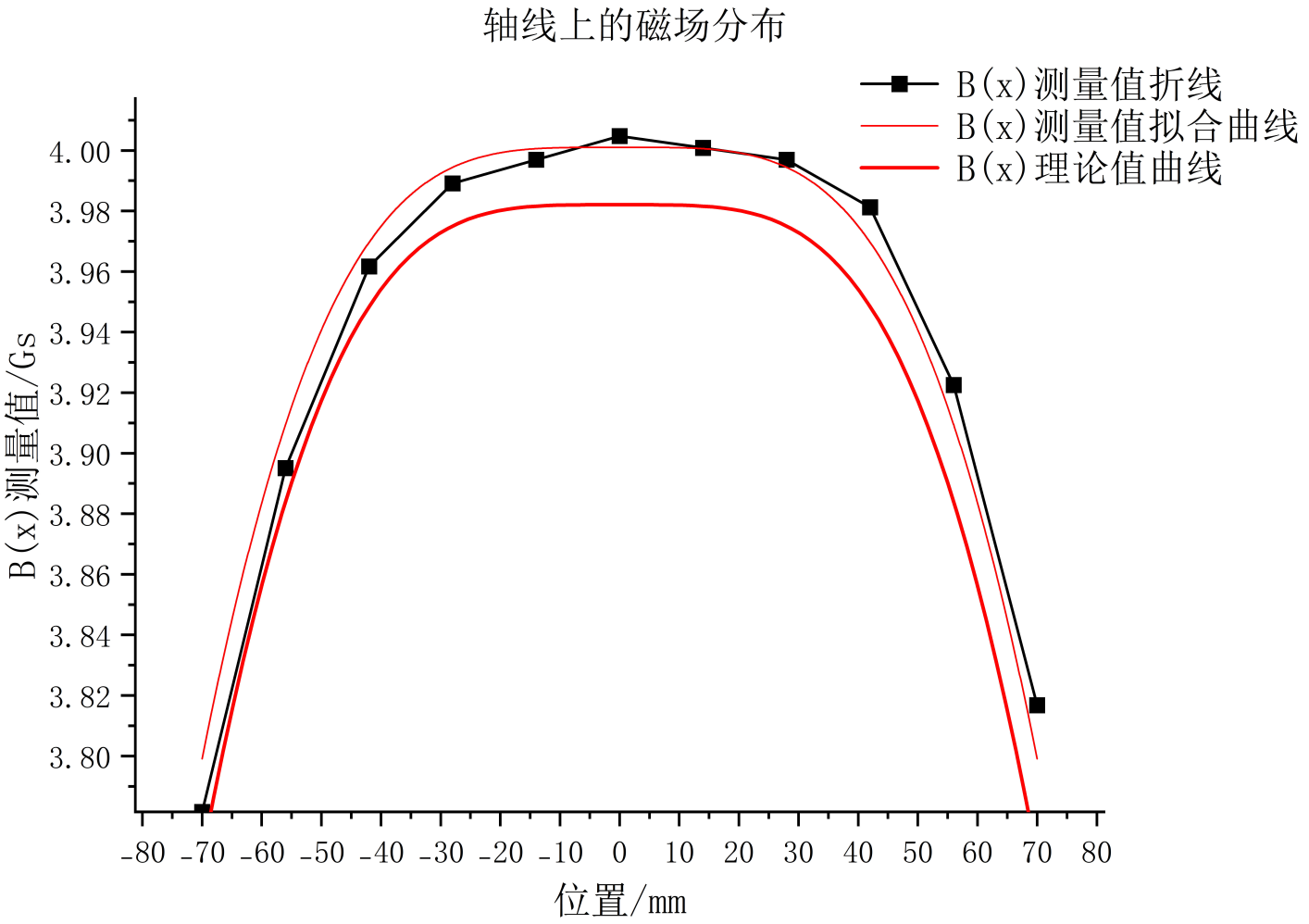
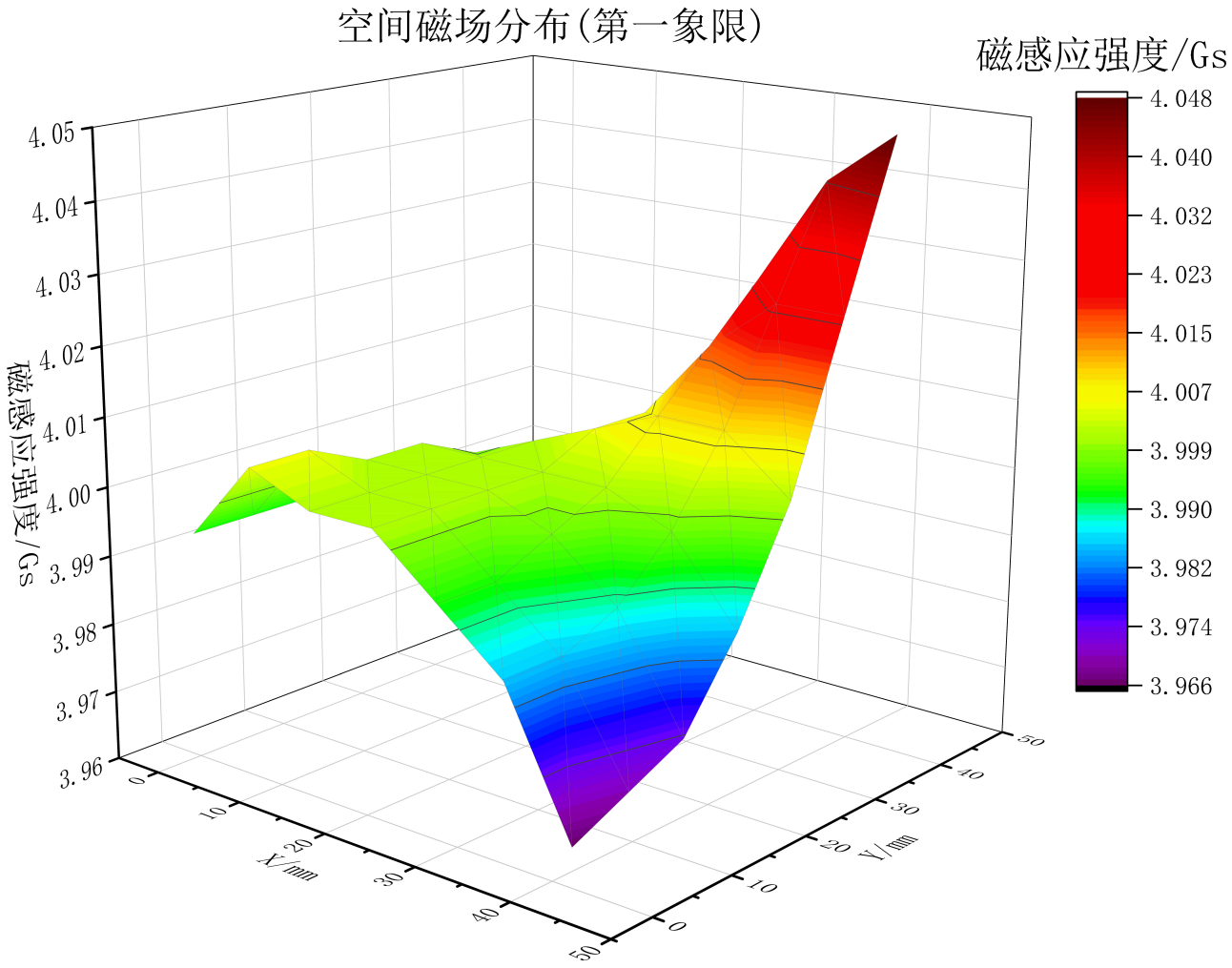
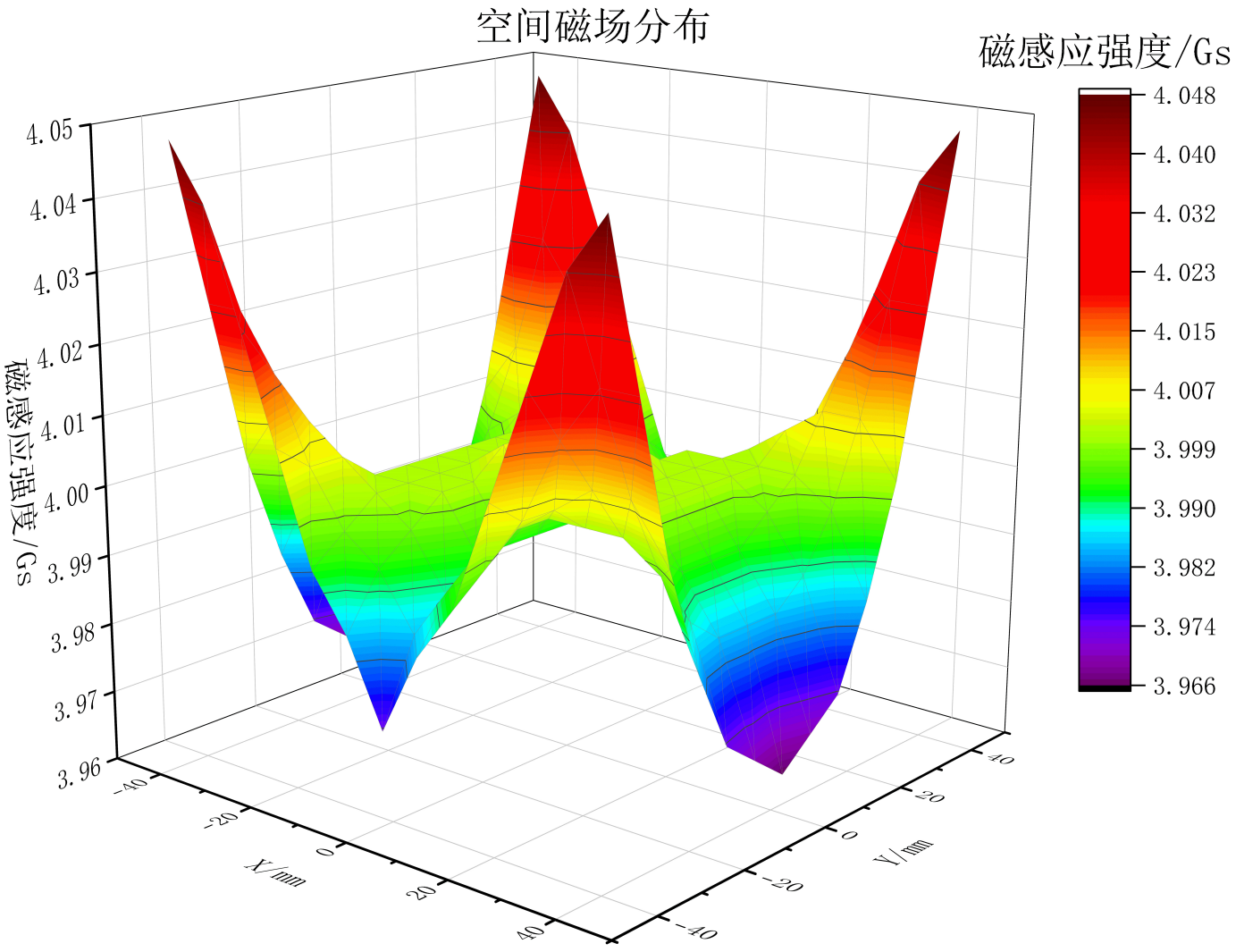

AMR的磁电转换特性曲线
方程 y = a + b*x
绘图 输出电压U
权重 不加权
截距 4.61538E-4 ± 0.004
斜率 0.25533 ± 0.00118
残差平方和 0.00276
Pearson's 0.99988
R平方(COD) 0.99977
调整后R平 0.99974
输出电压U/V
1
0
-1
-6
-4
-2
0
2
4
6
磁感应强度B/Gs


AMR的各向异性特性曲线
模型 Sine
方程 y=y0+A*sin(pi*(x-xc)/w)
绘图 输出电压U
y0 -0.31333 ± 0.05566
xc -103.2479 ± 3.11867
w 209.41034 ± 5.6698
A 1.33946 ± 0.05606
Reduced Chi-Sqr 1.91303E-6
R平方(COD) 0.99999
调整后R平方 0.99999
输出电压U/V
1.0
0.5
0.0
0
20
40
60
80
夹角θ/°


轴线上的磁场分布
B(x)测量值折线
B(x)测量值拟合曲线
B(x)理论值曲线
B(x)测量值/Gs
4.00
3.98
3.96
3.94
3.92
3.90
3.88
3.86
3.84
3.82
3.80
-80 -70 -60 -50 -40 -30 -20 -10 0 10 20 30 40 50 60 70 80
位置/mm


空间磁场分布(第一象限)
磁感应强度/Gs
4.048
4.040
4.032
4.023
4.015
4.007
3.999
3.990
3.982
3.974
3.966
磁感应强度/Gs
4.05
4.04
4.03
4.02
4.01
4.00
3.99
3.98
3.97
3.96
X/mm
Y/mm


空间磁场分布
磁感应强度/Gs
4.048
4.040
4.032
4.023
4.015
4.007
3.999
3.990
3.982
3.974
3.966
磁感应强度/Gs
4.05
4.04
4.03
4.02
4.01
4.00
3.99
3.98
3.97
3.96
-40
-20
0
20
40
X/mm
Y/mm